Institutional Cleansing and Political Scars:

Mafia-Related Municipal Dissolution, Electoral Participation, Political Supply, and Credit Disruption in Italy

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Abstract

This paper studies the political and economic consequences of mafia-related municipal dissolution in Italy. Dissolution is designed to restore legality after organised-crime infiltration of local government, yet it may also reveal institutional failure and disrupt the local infrastructure through which democratic participation is sustained. Combining municipality-level referendum and election data, annual snapshots of local administrators from 1985 to 2014, and SAFE–ORBIS evidence on firm credit conditions, we document three connected patterns. First, mafia-related dissolution is followed by a persistent decline in electoral participation. In the original referendum panel, treated municipalities experience a turnout decline of approximately 3.4 to 3.8 percentage points, and the effect remains negative under province-by-referendum-wave fixed effects. Modern staggered-adoption estimates are less precise, but the electorate-weighted group-time ATT remains negative and close to the province-by-wave fixed-effects estimate. Extending the electoral comparison beyond referenda shows that the decline is not limited to institutional or judicial referenda, although this extension is interpreted as diagnostic because election families differ in timing, salience, and national context. Second, the political-supply mechanism is strong. In annual administrator data, the preferred municipality and year fixed-effects specification, excluding 1985–1987 and the dissolution year, shows that dissolved municipalities have about 4.9 fewer administrators, 4.7 fewer councillors, and 7.3 percentage points higher administrator turnover after dissolution. Third, SAFE–ORBIS evidence indicates short-run credit-market spillovers. Firms in exposed provinces report worse bank-loan availability, lower confidence in banks, higher interest rates, and shorter loan maturities. These effects survive controls for contemporaneous firm fundamentals and are not reproduced by ordinary non-mafia municipal dissolutions. The evidence suggests that mafia-related dissolution does not only remove compromised administrations. It leaves a broad political participation scar and is accompanied by disruption in both local democratic infrastructure and local credit relationships.

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1 Introduction

What happens after the state publicly certifies that a local democratic institution has been infiltrated by organised crime? In Italy, mafia-related municipal dissolution is one of the strongest anti-mafia interventions available to the central state. When a municipal administration is deemed compromised by direct or indirect links to organised crime, elected local authorities are removed and replaced by an extraordinary commission. The intervention is meant to restore legality. Yet it also reveals a severe failure of local democratic representation. This paper asks whether this intervention leaves a political scar, and whether that scar is visible not only in voters' behaviour but also in the local political and economic infrastructure surrounding democratic participation.

The question matters because organised crime is not only a criminal phenomenon. It is a political and institutional phenomenon that can distort electoral competition, weaken public authority, and reshape the relationship between citizens and the state. Prior work shows that mafia presence has persistent effects on political competition, politician quality, public spending, procurement, firms, and local economic outcomes (Acemoglu et al., 2020; Acconcia et al., 2014; Daniele and Geys, 2015a,b; Fenizia, 2018; Galletta, 2016; Pinotti, 2015b). This paper shifts attention from mafia presence itself to the consequences of formal state intervention against mafia infiltration. The object is not only criminal capture, but what happens politically after the state exposes and removes a compromised local administration.

The paper began from a narrower hypothesis: mafia-related municipal dissolution might produce *status-quo demobilisation* in later institutional referenda. The original referendum analysis strongly supported persistent demobilisation and showed that, in the 2026 judicial-governance referendum, No votes as a share of the electorate fell more clearly than Yes votes rose. This remains an important composition result. However, the broader electoral evidence requires a more general interpretation. A comparison with non-referendum elections shows that the post-dissolution participation decline is not uniquely concentrated in judicial or institutional referenda. Instead, it appears across electoral families. We therefore frame the main claim as a broad political participation scar. Status-quo demobilisation is treated as a specific manifestation of this scar in the 2026 referendum and related institutional-referendum evidence, not as the only mechanism in the paper.

The central argument is therefore that mafia-related municipal dissolution disrupts the local infrastructures through which political participation is sustained. We use the word infrastructure in two senses. The first is political. Local administrators, councillors, mayors, party lists, and civic networks form the organisational environment that mobilises citizens and maintains democratic continuity. Dissolution directly interrupts this environment. The second is economic. Firms, banks, and local credit relationships form part of the broader institutional environment in which confidence in formal rules is produced or weakened. Dissolution may create uncertainty beyond the town hall, especially if it signals hidden criminal penetration of local economic and political networks.

The empirical analysis has three pillars. The first pillar is electoral. In the original municipality-by-referendum panel, municipalities exposed to mafia-related dissolution experience a turnout decline of approximately 3.4 to 3.8 percentage points. The effect remains negative when the specification includes municipality fixed effects and province-by-referendum-wave fixed effects. The 2026 composition evidence shows that No votes as a share of the electorate decline, while Yes votes as a share of the electorate do not rise robustly once weighting is introduced. A Southern matched DiD design reinforces the composition interpretation, showing a significant decline in No votes and a significant rise in Yes votes under exact-province matching, with no significant turnout change.

The new electoral-family extension modifies but strengthens the broader interpretation, while also requiring caution. When we compare institutional referenda with non-referendum elections, the decline is not stronger in judicial referenda than in other elections. Family-specific estimates show turnout effects of about -11.4 percentage points in non-referendum elections, -9.3 percentage points in the 2020 constitutional referendum, and -4.7 percentage points in the 2022 judicial referenda. A decomposition shows that the non-referendum estimate is driven primarily by the 2022 political election and the 2024 European election, while the 2020 regional election comparison is close to zero. We therefore use this extension as a diagnostic against a narrow referendum-only interpretation, not as the main estimate of the causal magnitude of dissolution across all elections.

The second pillar is political supply. We construct a municipality-year panel from annual snapshots of Italian local administrators between 1985 and 2014. The corrected panel contains 237,897 municipality-year observations and matches 271 treated municipalities. The results are strong. In the preferred specification, which excludes the less stable early years 1985–1987 and excludes the dissolution year itself, mafia-related dissolution is followed by about 4.9 fewer administrators, 4.7 fewer councillors, 0.7 fewer executive members, and a 7.3 percentage-point increase in administrator turnover. These results suggest that dissolution disrupts the local political class and the organisational channels through which turnout is normally mobilised.

The third pillar is credit-market spillovers. We merge SAFE survey data on firms' financing conditions with ORBIS balance-sheet data and assign exposure at the province-year level. This analysis is not a second municipality-level causal design, because SAFE–ORBIS identifies firm geography at the NUTS3/province level and SAFE is a survey rather than a census. It is therefore mechanism-consistent evidence of local economic spillovers. Still, the results are informative. Firms in exposed provinces report worse bank-loan availability, lower confidence in banks, higher interest rates, and shorter maturities in the first three years after exposure. These effects are not reproduced by ordinary non-mafia municipal dissolutions, and they remain in horse-race specifications that include both mafia-related and non-mafia dissolution exposure. At the same time, the evidence does not support a broad discouraged-borrower or rejection channel. The supported credit result is therefore narrower: credit conditions deteriorate at the intensive margin rather than through widespread application rejection or borrower exit from the credit market.

The paper makes four contributions. First, it extends the study of mafia-related municipal dissolution from local elections, public finance, procurement, and firm outcomes to national referenda and broader electoral participation. Second, it clarifies the composition of demobilisation by studying Yes and No votes relative to the full electorate. Third, it introduces direct municipal-level evidence on political supply after dissolution, showing that the local political class contracts and reshuffles. Fourth, it connects the political effects to firm-level evidence of short-run credit disruption, showing that mafia-related dissolution predicts worse credit conditions while ordinary non-mafia dissolutions do not.

The paper is careful about what it does not claim. The evidence does not prove individual-level distrust because the main electoral data are aggregate municipal returns. The identification hierarchy is explicit throughout the paper. The strongest causal evidence comes from municipality-level designs, especially the original referendum panel and the political-supply panel. The electoral-family comparison is a diagnostic extension because it compares different election types that occur in different years. The SAFE–ORBIS analysis is complementary spillover evidence because exposure is assigned at the province level and SAFE is a survey rather than a census. The political-supply evidence, while strong, partly captures the institutional mechanics of dissolution and is therefore interpreted as a mechanism of democratic disruption rather than as a separate behavioural outcome. Taken together, however, the evidence supports a cohesive interpretation: anti-mafia dissolution may be necessary to remove compromised administrations, but cleaning institutions is not the same as rebuilding democratic capacity.

The remainder of the paper is organised as follows. Section 3 describes mafia-related municipal dissolution and explains why it can be interpreted as an institutional intervention after local capture. Section 4 develops the conceptual framework. Section 5 describes the electoral, political-supply, SAFE–ORBIS, and non-mafia placebo data. Section 6 presents the empirical strategies. Sections 7, 8, and 9 report the electoral, political-supply, and SAFE results. Section 10 discusses interpretation and policy implications. Section 11 concludes.

2 Related Literature

This paper relates to five strands of literature. First, the empirical strategy builds on a research programme that exploits mafia-related municipal dissolution under Law 164/1991 and, subsequently, Article 143 TUEL as a quasi-experimental shock to local institutions. The foundational contribution is Acconcia et al. (2014), who use mafia-related dissolution as an instrument for unanticipated public spending contractions during commissioner administration and estimate local fiscal multipliers of approximately 1.5 to 1.9. Their paper develops the institutional and statistical case for treating dissolution timing as plausibly exogenous from the perspective of local jurisdictions and introduces the non-mafia dissolution placebo design used throughout this literature.

Subsequent work extends the same design to political and economic outcomes. Daniele and Geys (2015a) study the quality of subsequently elected politicians and show that mafia-related dissolution increases the average education of elected local politicians, especially mayors and aldermen. Daniele and Geys (2015b) study electoral outcomes and document that dissolution depresses voter turnout in subsequent local elections by 2.8 to 3.9 percentage points, with weaker spillovers to national-election turnout. Fenizia (2018) studies firm-level outcomes and shows that firms previously connected to dissolved administrations experience persistent reductions in their probability of winning public procurement tenders, consistent with stigma effects that outlast the commissioners' mandate. Galletta (2016) extends the design to local spending and spillover effects on neighbouring municipalities. This paper is closest to Daniele and Geys (2015b), but differs in both scope and mechanism. While they study turnout and party choice in elections, this paper studies referendum and election participation, referendum-side composition, local political supply, and firm-bank credit relationships after dissolution.

Second, the paper relates to work showing that organised crime and politically active armed groups distort democratic politics. Acemoglu et al. (2013) develop a probabilistic-voting model in which non-state armed actors use coercion and vote delivery to support politicians whose preferences align with theirs, creating a symbiotic equilibrium between armed actors and political elites. Acemoglu et al. (2020) study the historical emergence of the Sicilian mafia and document persistent effects on political competition, measured through vote concentration in parliamentary elections, over horizons of 60 to 80 years. This paper takes the existence of these political distortions as established and asks a downstream question: whether formal state intervention against mafia infiltration leaves persistent behavioural and organisational traces after the compromised administration has been removed.

Third, the paper contributes to the literature on organised crime, local institutions, and economic performance in Italy. Existing work shows that organised crime affects the quality of local institutions, the allocation of public resources, political selection, electoral outcomes, civic engagement, and social capital (Daniele and Marani, 2011; Buonanno et al., 2016; Daniele and Dipoppa, 2017; Daniele, 2019; Di Cataldo and Mastrorocco, 2022; Baraldi et al., 2022a,b; Buonanno et al., 2024). Pinotti (2015b) provides the canonical estimate of the economic cost of organised crime in Southern Italy, showing large and persistent losses in GDP per capita after mafia entry. Relative to this literature, the present paper shifts attention from the effects of mafia presence itself to the consequences of formal anti-mafia intervention, and it connects electoral behaviour to the local political and economic infrastructure surrounding democratic participation.

Fourth, the paper connects to the literature on political supply, political selection, and local democratic infrastructure. Political participation is not produced only by individual attitudes; it also depends on the parties, candidates, administrators, councillors, and civic networks that mobilise citizens and maintain democratic continuity. The political-supply results in this paper speak directly to this organisational margin. By showing that mafia-related dissolution is followed by fewer administrators, fewer councillors, and higher turnover, the paper provides evidence that the intervention disrupts the local political class through which participation is normally sustained. This mechanism complements the turnout results: demobilisation may arise not only because citizens change their beliefs, but also because the local political infrastructure that connects citizens to elections is weakened.

Fifth, the paper relates to work on institutional quality, trust, and credit relationships. A large literature argues that institutional quality and legal enforcement shape credit markets because banks and firms rely on predictable rules, enforceable contracts, and confidence in local economic relationships. Jappelli et al. (2005), for example, show that judicial enforcement affects credit markets. The SAFE–ORBIS evidence in this paper extends this logic to mafia-related institutional contamination. The results do not imply that credit disruption mechanically causes electoral demobilisation. Rather, they show that the same institutional shock that disrupts local democratic participation is also associated with short-run deterioration in firms' reported bank-loan availability, confidence in banks, interest rates, and loan maturities.

Finally, the paper connects to the literature on corruption, institutional trust, turnout, and selective demobilisation. A broader body of work argues that corruption, low institutional trust, and institutional failure tend to depress turnout by increasing cynicism, reducing civic duty, and weakening the perceived value of political participation (Stockemer et al., 2013; Sundström and Stockemer, 2015; Feitosa, 2020). It also relates to work on selective demobilisation and electoral composition. The idea that abstention is not politically neutral has a long tradition in political economy and electoral studies: different social and political groups display different mobilisation elasticities, and turnout changes may therefore alter the observable distribution of votes without changing underlying preferences in the full electorate (Lijphart, 1997; Verba et al., 1995). The contribution of this paper is to apply this logic to institutional referenda by defining status-quo demobilisation as a decline in *No* votes as a share of the electorate, while also showing that this composition result is part of a broader post-dissolution participation scar.

3 Institutional Background

3.1 Mafia-related municipal dissolution

Mafia-related municipal dissolution is an extraordinary intervention under Article 143 of the *Testo unico delle leggi sull'ordinamento degli enti locali* (TUEL). When a municipality is found to be subject to mafia conditioning or infiltration, the central state can dissolve the elected council and appoint an extraordinary commission. The commission temporarily replaces ordinary democratic authorities and is tasked with restoring legality and administrative functioning.

From the perspective of citizens, this is not a routine administrative event. It is a public certification that local democratic institutions have been compromised by organised crime. It may reveal that political representation, public procurement, administrative decisions, or local public services were shaped by actors outside the formal democratic process. The treatment is therefore both a legal intervention and an informational shock. It tells citizens that the local state failed.

This dual nature is central to the paper. If dissolution only removed corrupt or compromised officials and immediately restored confidence, one might expect participation to recover or even increase. If, instead, dissolution reveals that local institutions were deeply compromised and disrupts local political organisation, participation may decline. The empirical question is which of these forces dominates.

3.2 Dissolution as institutional intervention after local capture

The existing literature treats mafia-related dissolution as a useful institutional shock because dissolution is typically triggered by investigations, police evidence, or administrative findings of mafia infiltration rather than by ordinary electoral outcomes or local turnout trends (Acconcia et al., 2014; Daniele and Geys, 2015a,b). The design is not equivalent to random assignment, but the timing of dissolution is plausibly sharp relative to later referendum participation once municipality fixed effects, election-wave fixed effects, and geographic shocks are absorbed.

The intervention has several potential consequences. First, it removes the incumbent local political class. Second, it interrupts local networks linking administrators, parties, candidates, civic organisations, firms, and voters. Third, it creates stigma and uncertainty about the municipality. Fourth, it may change how citizens perceive institutions, legality, and the effectiveness of state action. These channels motivate the conceptual framework in the next section.

4 Conceptual Framework

4.1 Institutional revelation and political scars

Mafia-related dissolution is an institutional revelation. It signals that local democratic representation was not merely imperfect, but formally compromised by organised crime. Citizens may respond to this revelation in different ways. Some may become more supportive of state intervention and institutional reform. Others may become resigned, cynical, or less likely to participate. The aggregate outcome depends on which groups are mobilised or demobilised.

The paper defines a political scar as a persistent behavioural trace left by the revelation and removal of institutional capture. This scar is observed through lower participation in later elections and referenda. It does not require all citizens to lose trust. It only requires that the intervention changes the probability that some citizens participate, or changes the local networks that mobilise them.

4.2 Status-quo demobilisation and electoral composition

The referendum setting allows the paper to distinguish a decline in participation from a change in the composition of the active electorate. The key identity is

$$\frac{Yes_i}{Electorate_i} = \frac{Yes_i}{Valid_i} \times \frac{Valid_i}{Electorate_i}. \tag{1}$$

An increase in the Yes share among valid votes can therefore arise for two politically different reasons. It may reflect stronger reform mobilisation among the electorate, or it may reflect lower participation among voters who would otherwise have voted No.

For institutional and judicial referenda in which the No vote rejects institutional change, we define active status-quo voting as

$$SQ_i = \frac{No_i}{Electorate_i} \times 100. \tag{2}$$

A negative effect on SQ_i means that fewer citizens actively vote to preserve the existing institutional arrangement. In the broader version of the paper, status-quo demobilisation is not the only mechanism. It is a composition result nested inside a broader political-participation scar. The distinction remains important because it prevents the paper from interpreting a higher Yes share among valid votes as direct evidence of reform mobilisation when the same pattern may be generated by selective demobilisation of the status-quo side.

4.3 Political supply as democratic infrastructure

Political participation is not produced only by individual attitudes. It is also produced by local political supply. Administrators, councillors, mayors, civic lists, parties, and campaign networks provide information, mobilisation, and organisational continuity. They connect citizens to political choices. They also help transform preferences into turnout.

Dissolution can disrupt this infrastructure. By removing elected authorities and replacing them with commissioners, the state may weaken or reset local networks. In the short run, this can reduce the number of active political actors. In the medium run, it can increase turnover and alter the composition of the political class. If local mobilisation depends on these actors, then turnout can fall even without a direct change in citizens' preferences. This channel is especially important because the electoral-family results show broad demobilisation across electoral contexts.

4.4 Economic disruption and credit relationships

The third channel concerns local economic relationships. Mafia infiltration can affect firms, procurement, credit, and local confidence. Dissolution may reveal hidden institutional risk to banks and firms. Even if the intervention is legally corrective, it may create uncertainty about the quality of local networks, the reliability of business relationships, and the risk environment surrounding firms in exposed areas.

The SAFE–ORBIS analysis is designed to capture this dimension as a parallel spillover of the same institutional shock. It does not claim that credit disruption mechanically causes turnout decline. Instead, it asks whether the same institutional shock that disrupts local politics is also visible in firm-bank relationships. Evidence of worse credit conditions, higher interest rates, shorter maturities, and lower bank-facing confidence supports the broader interpretation of dissolution as a local institutional shock that unsettles both political and economic infrastructure.

5 Data

5.1 Municipal dissolution and referendum/election data

The main treatment variable identifies municipalities exposed to mafia-related dissolution before the relevant election or referendum. In the original referendum panel, the unit of observation is the municipality-by-referendum wave. The panel covers seven national referendum waves: 2009, 2011, April 2016, December 2016, 2020, 2022, and 2026. The final panel contains 54,970 municipality-wave observations and 7,894 municipalities. Treatment is concentrated in Southern Italy, and identification comes from municipalities that switch into treatment during the observation window.

The 2026 cross-sectional composition analysis uses municipality-level results from the judicial-governance constitutional referendum. The returns are taken from the Ministry of the Interior's Eligendo municipal results files and should be read as final official municipal returns as downloaded in June 2026. The main outcomes are Yes votes as a share of the electorate, No votes as a share of the electorate, valid votes as a share of the electorate, and the Yes share among valid votes. The key status-quo outcome is No votes as a share of the electorate because the No vote defended the existing institutional arrangement.

The extended electoral-family panel adds comparison elections. It includes the 2020 constitutional referendum, the five 2022 judicial referenda, the 2022 political election, the 2024 European election, and 2020 regional elections in regions that voted in 2020. After aggregating the five 2022 judicial referendum questions to a unique municipality-election turnout panel, the extension contains 33,207 municipality-election observations. This extension is used to test whether the post-dissolution turnout decline is specific to institutional or judicial referenda or instead reflects a broader decline in political participation.

5.2 Local administrators and political supply

To study political supply, we use annual snapshots of Italian municipal administrators from the historical local-administrator archive covering 1985–2014. Each snapshot contains the individuals serving in municipal offices at the end of the year, including mayors, councillors, executive members, and other administrators. We parse the semicolon-separated municipal files, harmonise municipality and province names, and build a municipality-year panel.

The corrected political-supply panel contains 237,897 municipality-year observations. From 1988 onward, coverage is stable at roughly 8,000 municipalities per year. The panel matches 271 treated municipalities, close to the 274 treated municipalities identified in the treatment data. The main outcomes are the number of administrators, number of unique persons, number of councillors, number of executive members, administrator turnover relative to the previous annual snapshot, mayor change relative to the previous annual snapshot, female share, mean age, and the number of distinct professions among administrators.

The political-supply data end in 2014. This means that the mechanism analysis applies to dissolutions observed before or by 2014 and cannot evaluate later treatment events. The preferred specification excludes 1985–1987 because early coverage is less stable, and it excludes the dissolution year itself because the immediate intervention year mechanically reflects the replacement of elected authorities by commissioners.

5.3 SAFE–ORBIS firm-credit data

The SAFE–ORBIS analysis combines the Survey on the Access to Finance of Enterprises with ORBIS balance-sheet data. SAFE provides survey information on firms' financing conditions, while ORBIS adds firm characteristics and geographic information. The province-year analysis covers 2014–2024 and contains 1,136 province-year cells, 105 provinces, and 25,722 underlying firm-wave observations.

The key limitation is geographic resolution. Firm location is observed at the NUTS3 level, corresponding broadly to provinces, not at the municipality level. We therefore define exposure at the province-year level. A province becomes exposed when at least one municipality in the province has experienced mafia-related dissolution. The SAFE–ORBIS analysis should therefore be interpreted as credit-market spillover evidence, not as a municipality-level causal design.

The main SAFE outcomes measure worsened bank-loan availability, worsened credit-line availability, low confidence talking with banks about financing, higher interest rates, decreased available loan size, decreased loan maturity, increased collateral requirements, discouraged borrowing, and bank-loan rejection. Non-applicable, refusal, and don't-know responses are treated as missing. SAFE survey weights are used when available. The final results show that the strongest supported outcomes are worsened bank-loan availability, lower confidence in banks, higher interest rates, and shorter maturities.

5.4 Official non-mafia dissolution placebo data

To distinguish mafia-related institutional contamination from ordinary municipal instability, we construct an official non-mafia municipal-dissolution placebo from the Ministry of Interior INT00063 files on dissolutions of municipal and provincial councils. The placebo file covers 2016–2024 and contains 1,118 ordinary non-mafia municipal dissolutions after excluding rows referring to mafia infiltration, conditioning, or Article 143. The retained reasons include ordinary political and administrative causes such as councillor resignations, mayor resignations, mayor death, no-confidence votes, and other non-mafia grounds for dissolution.

The non-mafia placebo is matched to the SAFE province-year panel by province name. The match rate is 98.2 percent, with 1,098 of 1,118 dissolution rows matching a SAFE province. We use this file to construct province-year placebo exposure indicators. The main placebo equals one in the first three years after the first ordinary non-mafia dissolution in a province. This mirrors the mafia-related short-run exposure window used in the SAFE–ORBIS analysis.

6 Empirical Strategy

6.1 Hierarchy of evidence and identifying assumptions

The paper combines three empirical pillars, but they do not have the same inferential status. Table 1 makes this hierarchy explicit. The municipality-level electoral and political-supply designs provide the strongest evidence because they exploit within-municipality variation before and after mafia-related dissolution. The electoral-family comparison is useful for interpretation, but it is diagnostic rather than the cleanest causal design because election families occur in different years and differ in salience, national context, and electorate-level mobilisation. The SAFE–ORBIS analysis is credit-market spillover evidence at the province level. It shows whether the same institutional shock is associated with changes in firms' reported credit conditions, but it does not identify firm-level exposure inside dissolved municipalities.

Table 1: Hierarchy of evidence

Empirical component	Unit and design	Inferential status	Main identifying assumption and use in the paper
Original referendum panel	Municipality by referendum wave, municipality FE and wave or province-by-wave FE	Main causal evidence	In the absence of dissolution, treated municipalities would have followed similar within-province turnout changes across referendum waves. Used to establish the electoral participation scar.
2026 referendum composition	Municipality cross-section with province FE, controls, matching, and permutation checks	Mechanism-oriented electoral evidence	Conditional on geography and observables, treated and comparison municipalities provide informative contrasts in vote composition. Used to study status-quo demobilisation, not to replace the panel design.
Electoral-family extension	Municipality by election family, municipality FE and election FE	Diagnostic extension	Useful for showing whether the turnout scar is referendum-specific, but vulnerable to election-type timing and salience. Used to broaden interpretation, not as the strongest causal result.
Political-supply panel	Municipality by year, municipality FE and year FE	Main mechanism evidence	In the absence of dissolution, treated municipalities would have followed similar changes in local political supply. The preferred sample excludes 1985–1987 and the dissolution year to reduce coverage and mechanical concerns.
SAFE–ORBIS	Province by SAFE wave, province-level exposure, province FE and year FE	Credit-market spillover evidence	Conditional on province and year effects and lagged ORBIS characteristics, exposed province-years reveal whether credit

The identifying assumptions differ by pillar. For the electoral panel, the key condition is parallel within-municipality turnout trends after absorbing referendum-wave shocks and, in the stricter specification, province-by-wave shocks. For the political-supply panel, the key condition is parallel changes in administrator outcomes after absorbing municipality and year fixed effects. For SAFE–ORBIS, the claim is weaker: the design tests whether province-level exposure coincides with worse credit conditions among surveyed firms after controlling for province and year effects and lagged ORBIS characteristics. The SAFE results therefore support the broader interpretation of local institutional disruption, but they are not presented as proof that credit tightening causes the electoral scar.

6.2 Electoral participation design

The baseline referendum turnout specification is

$$T_{it} = \alpha_i + \lambda_t + \beta Post_{it} + \varepsilon_{it}, \tag{3}$$

where T_{it} is turnout in municipality i during referendum wave t, α_i are municipality fixed effects, λ_t are referendum-wave fixed effects, and $Post_{it}$ equals one once municipality i has experienced mafia-related dissolution. Standard errors are clustered at the province level.

Because mafia-related dissolution is geographically concentrated, we also estimate a stricter specification with province-by-wave fixed effects:

$$T_{it} = \alpha_i + \lambda_{p(i)t} + \beta Post_{it} + \varepsilon_{it}. \tag{4}$$

This absorbs shocks common to municipalities in the same province during the same referendum wave.

The 2026 composition analysis estimates

$$y_i = \alpha + \beta D_i + X_i' \gamma + \delta_{p(i)} + \varepsilon_i, \tag{5}$$

where y_i is one of the electorate-based vote outcomes, D_i indicates prior mafia-related dissolution, X_i includes municipal controls, and $\delta_{p(i)}$ are province fixed effects. The key outcome is No votes as a share of the electorate.

The electoral-family extension estimates family-specific post-dissolution effects:

$$T_{ie} = \alpha_i + \lambda_e + \beta_1 (Post_i \times NonRef_e) + \beta_2 (Post_i \times ConstRef_e) + \beta_3 (Post_i \times JudRef_e) + u_{ie}, \tag{6}$$

where e indexes election families, $NonRef_e$ identifies non-referendum elections, $ConstRef_e$ identifies the 2020 constitutional referendum, and $JudRef_e$ identifies the 2022 judicial referenda. This specification avoids interpreting a main effect plus interactions and gives one coefficient per election family.

Because treatment adoption is staggered across municipalities, we also report modern staggered-adoption diagnostics for the main referendum turnout result. These diagnostics are motivated by concerns that two-way fixed-effects estimators may be difficult to interpret when treatment effects are heterogeneous across cohorts or event time (Sun and Abraham, 2021; Callaway and Sant'Anna, 2021; Borusyak et al., 2024; de Chaisemartin and D'Haultfoeuille, 2020). We implement a Callaway–Sant'Anna-style group-time estimator that compares newly treated municipalities with never-treated and not-yet-treated municipalities and aggregates group-time average treatment effects over post-treatment referendum waves. This estimator uses only municipalities treated within the referendum panel and with an observed pre-treatment referendum wave, so its estimand differs from the TWFE estimand. We therefore use it as a robustness and cautionary diagnostic rather than as the main estimator.

6.3 Political-supply design

The political-supply mechanism uses municipality-by-year administrator data. The baseline specification is

$$Y_{it} = \alpha_i + \lambda_t + \beta PostDissolution_{it} + \varepsilon_{it}, \tag{7}$$

where Y_{it} is a political-supply outcome in municipality i and year t, α_i are municipality fixed effects, and λ_t are year fixed effects. Standard errors are clustered at the municipality level. The preferred specification restricts the sample to 1988–2014 and excludes the dissolution year.

The political-supply design requires special care because the intervention itself removes elected authorities. A fall in administrators during the dissolution year is therefore partly mechanical. For this reason, we report the dissolution-year exclusion as the preferred specification and interpret the remaining post-treatment coefficients as evidence of disruption beyond the immediate legal replacement of the council. To push this point further, the paper defines persistence bins around first dissolution:

$$Y_{it} = \alpha_i + \lambda_t + \beta_{1-3}\mathbf{1}\{1 \leq t - G_i \leq 3\} + \beta_{4-6}\mathbf{1}\{4 \leq t - G_i \leq 6\} + \beta_{7+}\mathbf{1}\{t - G_i \geq 7\} + \varepsilon_{it}, \tag{8}$$

where G_i is the first dissolution year. The omitted treated period is the year immediately before dissolution, while never-treated municipalities contribute to the fixed-effects comparison throughout the sample. This specification separates the short-run post-commissioner period from medium-run and longer-run persistence.

We also estimate event-time specifications that bin years relative to first dissolution. These are interpreted as timing diagnostics rather than as the main causal design because some pre-treatment coefficients are not perfectly flat. The event-time figure is therefore used to show the timing of the contraction and reshuffling, while the preferred post-1988 specification excluding the dissolution year remains the main mechanism table.

6.4 SAFE–ORBIS credit-spillover design

The SAFE–ORBIS specification is estimated at the province-year level:

$$Y_{pt} = \beta D_{pt}^{0-3} + X'_{p,t-1}\gamma + \alpha_p + \lambda_t + \varepsilon_{pt}, \tag{9}$$

where Y_{pt} is a province-year SAFE outcome in province p and year t, D_{pt}^{0-3} equals one in the first three years after the first mafia-related dissolution in province p, $X_{p,t-1}$ includes province-year averages of lagged ORBIS controls, α_p are province fixed effects, and λ_t are year fixed effects. Standard errors are clustered at the NUTS3 province level.

The control vector includes lagged firm size, employment, profitability, leverage, liquidity, and solvency. The SAFE evidence is not interpreted as a standalone municipality-level causal design because exposure is assigned at the province level and SAFE is a survey. Instead, it tests whether the same institutional shock is associated with short-run deterioration in firm-bank credit conditions.

The main placebo design replaces mafia-related exposure with ordinary non-mafia municipal dissolution. Let N_{pt}^{0-3} equal one in the first three years after the first non-mafia municipal dissolution in province p. We estimate

$$Y_{pt} = \beta_N N_{pt}^{0-3} + X'_{p,t-1}\gamma + \alpha_p + \lambda_t + \varepsilon_{pt}. \tag{10}$$

The horse-race specification includes both treatment variables:

$$Y_{pt} = \beta_M D_{pt}^{0-3} + \beta_N N_{pt}^{0-3} + X'_{p,t-1}\gamma + \alpha_p + \lambda_t + \varepsilon_{pt}. \tag{11}$$

If the SAFE results reflected generic administrative disruption, the non-mafia placebo should produce similar effects. If instead the results are tied to mafia-related institutional contamination, β_M should remain positive for the credit-condition outcomes while β_N should be small and statistically insignificant.

Because SAFE exposure is defined at the province-year level, fully saturated province-by-year fixed effects are not feasible in the SAFE–ORBIS specification. They absorb the treatment variation by construction. We therefore use province fixed effects and year fixed effects as the headline specification and interpret the SAFE evidence as province-year credit-market spillover evidence rather than as a municipality-level causal design.

Credit-vulnerability diagnostics. To assess whether the SAFE–ORBIS results are driven by firms that were already vulnerable to credit tightening, we construct predicted credit-vulnerability scores. For each credit dimension k, we estimate a prediction model using only never-exposed and pre-exposure firm-wave observations:

$$\Pr(V_{ipt}^k = 1) = F\left(Z'_{i,p,t-1}\pi + \rho_p + \tau_t\right),$$

where V_{ipt}^k denotes a credit-vulnerability outcome, $Z_{i,p,t-1}$ contains lagged ORBIS firm fundamentals, and ρ_p and τ_t capture geographic and time predictors. We consider four vulnerability dimensions: availability risk, price-and-term risk, hard-rationing risk, and bank-confidence risk. The fitted value $\widehat{p}_{ipt}^k$ is then used to test whether the post-dissolution SAFE effects vary with ex ante credit vulnerability.

The main diagnostic specification is

$$Y_{ipt} = \beta D_{pt}^{0-3} + \theta\left(D_{pt}^{0-3} \times \widetilde{p}_{ipt}^k\right) + \delta \widetilde{p}_{ipt}^k + X'_{i,p,t-1}\gamma + \alpha_p + \lambda_t + \varepsilon_{ipt},$$

where $\widetilde{p}_{ipt}^k$ is the predicted vulnerability score centered at its median and scaled by ten percentage points. Thus, β measures the exposure effect for a firm at median predicted vulnerability, while θ measures how the effect changes when predicted vulnerability rises by ten percentage points. As a complementary split-sample diagnostic, we also compare firms below and above the median of $\widehat{p}^{k}_{ipt}$.

7 Electoral Participation Results

7.1 Main referendum evidence

Table 2 reports the original referendum-panel estimates. Mafia-related dissolution is followed by a significant decline in referendum turnout. The unweighted estimate is −3.37 percentage points, while the electorate-weighted estimate is −3.80 percentage points.

Table 2: Effect of mafia-related dissolution on referendum turnout

	(1)	(2)
Post-dissolution	-3.368^{***}	-3.802^{**}
	(0.957)	(1.641)
Municipality FE	✓	✓
Referendum-wave FE	✓	✓
Weighting	None	Electorate
Observations	54,970	54,970
Municipalities	7,894	7,894
Province clusters	110	110

Notes: The dependent variable is referendum turnout in percentage points. Standard errors are clustered at the province level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 3 shows that the effect remains negative under province-by-wave fixed effects. The stricter unweighted estimate is −1.81 percentage points and the electorate-weighted estimate is −2.58 percentage points. This indicates that the result is not driven only by province-specific shocks in particular referendum waves.

Table 3: Fixed-effects robustness for referendum turnout

	(1)	(2)	(3)	(4)
Post-dissolution	-3.368^{***}	-3.802^{**}	-1.813^{***}	-2.579^{**}
	(0.957)	(1.641)	(0.569)	(1.006)
Municipality FE	✓	✓	✓	✓
Wave FE	✓	✓		
Province × Wave FE			✓	✓
Weighting	None	Electorate	None	Electorate
Observations	54,970	54,970	54,970	54,970

Notes: The dependent variable is referendum turnout in percentage points. Standard errors are clustered at the province level.

The original event-study diagnostics do not show strong evidence of adverse pre-trends. Post-treatment coefficients are generally negative, with the largest decline around three referendum waves after treatment. Table 4 reports modern staggered-adoption diagnostics for the same main referendum turnout outcome. The TWFE estimates remain negative and statistically significant. The municipality and wave fixed-effects estimate is -3.37 percentage points, while the stricter province-by-wave fixed-effects estimate is -1.81 percentage points. The Callaway–Sant'Anna-style group-time estimates are less precise. The unweighted 0–3 wave ATT is -0.68 percentage points and statistically insignificant. The electorate-weighted 0–3 wave ATT is -2.00 percentage points, close to the province-by-wave fixed-effects estimate, but also statistically insignificant. We therefore interpret the staggered-adoption exercise as a robustness and cautionary diagnostic. It supports the negative direction of the participation scar, especially in electorate-weighted specifications, while indicating that the exact magnitude should not be over-interpreted.

Table 4: Staggered-adoption robustness for referendum turnout

Estimator	Coefficient	Standard error	p-value
TWFE, wave FE	-3.368^{***}	(1.035)	0.001
TWFE, province × wave FE	-1.813^{***}	(0.620)	0.003
CS-style ATT, 0–3 waves, unweighted	-0.683	(1.015)	0.501
CS-style ATT, 0–3 waves, electorate weighted	-2.001	(1.840)	0.277
CS-style ATT, 1–3 waves, electorate weighted	-2.355	(1.894)	0.214

Notes: The dependent variable is referendum turnout in percentage points. TWFE specifications include municipality fixed effects and either referendum-wave fixed effects or province-by-wave fixed effects. CS-style estimates are group-time ATT estimates that use not-yet-treated and never-treated municipalities as controls. Standard errors for TWFE estimates are clustered by province. Standard errors for CS-style estimates are obtained by province bootstrap with 300 replications. The CS-style estimates use only municipalities treated within the referendum panel and with an observed pre-treatment referendum wave, so the estimand differs from the TWFE estimand. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

7.2 Referendum-frame heterogeneity and negative-control frames

Before turning to the broader electoral-family extension, we first revisit heterogeneity inside the original referendum panel. This exercise is useful because it asks whether the referendum turnout penalty is concentrated in referendum frames that are substantively close to the institutional shock created by mafia-related dissolution. Table 5 reports strict frame-specific estimates with municipality fixed effects and province-by-referendum-wave fixed effects. The turnout penalty is concentrated in judicial and legal-institutional referenda. Institutional non-judicial referenda and all non-judicial referenda do not display comparable negative effects.

Table 5: Negative-control referendum frames: strict turnout effects

	(1)	(2)
Judicial/legal referenda	-3.017^{***}	-3.935^{***}
	(0.829)	(1.329)
Institutional non-judicial referenda	-0.505	0.170
	(1.001)	(2.703)
All non-judicial referenda	-0.507	-0.506
	(0.950)	(2.359)
Municipality FE	✓	✓
Province × Wave FE	✓	✓
Weighting	None	Electorate

Notes: The dependent variable is referendum turnout in percentage points. Each coefficient reports the post-dissolution turnout effect estimated separately within the corresponding referendum frame. Standard errors are clustered at the province level. All specifications include municipality fixed effects and province-by-wave fixed effects. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

This evidence should be interpreted together with the electoral-family extension below. Within referenda, the demobilisation effect is strongest in judicial and legal-institutional frames. Once non-referendum elections are added, however, the participation scar is not limited to judicial or institutional referenda. The paper therefore keeps the frame result as referendum-internal evidence on where status-quo demobilisation is most visible, while using the electoral-family extension to broaden the participation interpretation.

7.3 Electoral-family comparison

The electoral-family extension asks whether the turnout decline is specific to referenda. This exercise is explicitly diagnostic. It pools elections with different national contexts, salience, timing, and campaign environments, so it should not be read as the cleanest causal design in the paper. Its purpose is narrower: it tests whether the post-dissolution turnout scar appears only when voters face institutional or judicial referenda, or whether it is visible across electoral contexts. Table 6 reports family-specific effects. The answer is that the turnout decline is not referendum-specific. The estimated effect is largest in non-referendum elections, followed by the 2020 constitutional referendum, and then the 2022 judicial referenda. Because the non-referendum coefficient is much larger than the original referendum-panel estimate, we report election-specific diagnostics below and treat the family comparison as a robustness and interpretation exercise rather than as the primary magnitude estimate.

Table 6: Family-specific turnout effects

Election family	Coefficient	Standard error	p-value
Non-referendum elections	-11.426^{***}	(2.566)	< 0.001
2020 constitutional referendum	-9.315^{***}	(2.823)	0.001
2022 judicial referenda	-4.729^{*}	(2.798)	0.091
Municipality FE	✓		
Election FE	✓		
Observations	33,207		

Notes: The dependent variable is turnout. Coefficients are expressed in percentage points. The specification includes municipality and election fixed effects and clusters standard errors by municipality. Non-referendum elections include the 2022 political election, the 2024 European election, and 2020 regional elections in voting regions. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

This result clarifies the interpretation of the paper. The evidence does not support the claim that judicial or institutional referenda are uniquely affected. It supports a broader political-participation scar. Mafia-related dissolution appears to weaken participation across electoral contexts. Institutional and judicial referenda remain important because they reveal how demobilisation changes vote composition, but the participation decline itself is broader. At the same time, the family comparison is not a substitute for the original referendum panel. It combines election types observed in different years and under different salience, turnout baselines, campaign environments, and national political contexts. We therefore use the electoral-family extension to reject a narrow referendum-only interpretation, not to claim that ordinary elections are necessarily more causally affected than referenda or that the coefficients are directly comparable across election families.

Table 7 decomposes the family comparison by election. The large non-referendum estimate is not driven by the 2020 regional election, which is close to zero. It is driven by the two national non-referendum elections in the extension, namely the 2022 political election and the 2024 European election. This reinforces the diagnostic interpretation: the extension shows that the participation scar is visible outside referenda, but it also shows that the magnitude is sensitive to which comparison elections are included.

Table 7: Election-specific turnout diagnostics

Election	Coefficient	Standard error	p-value
2020 constitutional referendum	-8.426^{***}	(2.900)	0.004
2020 regional elections	-0.293	(2.855)	0.918
2022 judicial referenda	-3.901	(2.843)	0.170
2022 political election	-12.624^{***}	(2.666)	< 0.001
2024 European election	-11.574^{***}	(2.597)	< 0.001

Notes: The dependent variable is turnout. Coefficients are expressed in percentage points. The model includes municipality and election fixed effects and clusters standard errors by municipality. This table decomposes the diagnostic family comparison by election.

7.4 Vote composition and status-quo mobilisation

The 2026 composition analysis remains important because it shows how demobilisation is distributed across referendum sides. Table 8 reports the central decomposition. No votes as a share of the electorate decline significantly, while Yes votes as a share of the electorate do not increase robustly once weighting is introduced. Since No defended the existing judicial-institutional arrangement, this is evidence of status-quo demobilisation in the 2026 referendum.

Table 8: 2026 referendum decomposition

	Yes votes / electorate	No votes / electorate	Valid votes / electorate
Panel A: Unweighted			
Mafia-related dissolution	1.094^{*}	-2.067^{***}	-0.972^{***}
	(0.627)	(0.468)	(0.357)
Panel B: Weighted by electorate			
Mafia-related dissolution	0.119	-1.110^{**}	-0.991^{***}
	(0.311)	(0.441)	(0.312)
Province FE	✓	✓	✓
Controls	✓	✓	✓
Observations	7,890	7,890	7,890

Notes: Coefficients are expressed in percentage points. Standard errors are clustered at the province level. In the 2026 judicial-governance referendum, No votes are interpreted as active status-quo votes.

A Southern matched DiD design sharpens the composition result but uses different identifying variation. Under exact-province matching, No votes as a share of the electorate fall by 1.24 percentage points, Yes votes as a share of the electorate rise by 1.41 percentage points, and turnout does not change significantly. This points to compositional reallocation within the electorate rather than pure abstention.

The full-sample 2026 specification and the Southern matched DiD should therefore be read as complementary rather than identical. The full-sample cross-section uses national municipal variation with province fixed effects and controls. It shows that the fall in valid participation is concentrated on the No side: No votes decline, Yes votes do not rise robustly once weighting is introduced, and valid votes decline. The Southern matched DiD uses a narrower comparison set and exact-province matching. It shows a decline in No votes together with an increase in Yes votes and no significant turnout change. Together, the two exercises indicate that status-quo support weakens after dissolution, but they do not identify a single uniform behavioural margin across all municipalities. In some settings, the pattern looks like status-quo abstention. In the matched Southern comparison, it looks more like compositional reallocation.

The 2026 result also survives a direct control for revealed pre-2026 judicial-referendum behaviour. Table 9 controls for the municipality-level average across the 2022 justice referenda in No votes as a share of the electorate and valid votes as a share of the electorate. These controls absorb pre-existing differences in judicial-referendum status-quo voting and participation. The No/electorate coefficient remains negative and statistically significant in both the unweighted and electorate-weighted specifications, while Yes/electorate remains statistically insignificant.

Table 9: 2026 composition effects conditioning on 2022 justice-referendum behaviour

Outcome	Weighting	Coefficient	Standard error	Holm p-value
Yes votes / electorate	None	0.535	0.505	0.290
No votes / electorate	None	−1.458	0.344	0.0001
Valid votes / electorate	None	−0.923	0.352	0.0297
Yes votes / electorate	Electorate	0.031	0.293	0.916
No votes / electorate	Electorate	−1.014	0.334	0.0072
Valid votes / electorate	Electorate	−0.983	0.299	0.0041

Notes: The table reports 2026 composition regressions that condition on pre-2026 judicial-referendum behaviour. The proxies are the municipality-level averages across the 2022 justice referenda for No votes as a share of the electorate and valid votes as a share of the electorate. All specifications include province fixed effects and the baseline municipal controls. Standard errors are clustered at the province level. Holm p-values adjust across the three primary electorate-based outcomes within each weighting scheme.

The matched Southern DiD evidence provides an additional composition check. Table 10 reports average treatment effects on the treated using exact-province matching. No votes as a share of the electorate fall significantly, Yes votes as a share of the electorate rise significantly, and turnout does not change significantly. This pattern is more consistent with compositional reallocation within the active electorate than with pure turnout decline.

Table 10: Southern matched DiD for 2026 composition outcomes

Outcome	ATT	Standard error	p-value
Yes votes / electorate	1.409^{***}	(0.460)	0.0026
No votes / electorate	-1.239^{***}	(0.386)	0.0016
Valid votes / electorate	0.235	(0.671)	0.7269
Yes share / valid votes	1.372^{***}	(0.405)	0.0009
Turnout	0.407	(0.893)	0.6491

Notes: The table reports average treatment effects on the treated from a South-only matched DiD design. The outcome is the change from the pre-2026 mean based on 2020 and 2022 referendum-composition data to the 2026 referendum. Treated municipalities are matched to five nearest untreated neighbours with exact province matching. Matching features include the pre-period value of the relevant outcome and the standard socio-demographic covariates. The matched sample contains 170 treated municipalities and 559 available controls. Coefficients are expressed in percentage points. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 11 compares the five pooled 2022 justice referenda with the 2020 constitutional referendum. The earlier-referendum evidence is consistent with a cautious interpretation. In 2022, treated municipalities display a higher Yes share among valid votes, but this comes together with declines in both Yes and No votes as shares of the electorate, with the No side falling more. By contrast, the 2020 constitutional referendum does not display a comparable electorate-based pattern. The 2026 result is therefore the clearest status-quo demobilisation result, while the

2022 justice evidence is supportive but less clean and the 2020 benchmark is not comparable.

Table 11: Composition evidence from earlier referenda

	Yes votes / valid votes	Yes votes / electorate	No votes / electorate	Valid votes / electorate
Panel A: 2022 justice referenda				
Mafia-related dissolution	2.40^{***}	-0.92^{**}	-1.32^{***}	-2.24^{***}
p-value	[0.0004]	[0.0320]	[0.0008]	[0.0040]
Province FE	✓	✓	✓	✓
Question FE	✓	✓	✓	✓
Controls	✓	✓	✓	✓
Panel B: 2020 constitutional referendum				
Mafia-related dissolution	0.13	-0.32	-0.11	-0.43
p-value	[0.7530]	[0.5620]	[0.6340]	[0.4900]
Province FE	✓	✓	✓	✓
Controls	✓	✓	✓	✓

Notes: Each coefficient reports the effect of mafia-related municipal dissolution on the indicated referendum-composition outcome. Brackets report p-values. The 2022 estimates pool the five justice-related referendum questions and include province fixed effects, question fixed effects, and the standard municipal controls. The 2020 estimates use the same municipality-level decomposition logic for the constitutional referendum and include province fixed effects and controls. Coefficients are expressed in percentage points. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

8 Political-Supply Disruption

8.1 Contraction of local administrators and councillors

The political-supply evidence is the strongest mechanism result. Table 12 reports the preferred specification, which excludes 1985–1987 and excludes the dissolution year. Mafia-related dissolution is followed by a large contraction in the local political-administrative class. Treated municipalities have about 4.86 fewer administrators, 4.66 fewer councillors, and 0.67 fewer executive members after dissolution.

Table 12: Political-supply effects of mafia-related municipal dissolution

Outcome	Coefficient	Standard error	p-value
Number of administrators	-4.858^{***}	(0.377)	< 0.001
Number of councillors	-4.662^{***}	(0.343)	< 0.001
Number of executive members	-0.672^{***}	(0.109)	< 0.001
Administrator turnover rate	7.264^{***}	(1.129)	< 0.001
Distinct professions	-0.901^{***}	(0.225)	< 0.001
Municipality FE	✓		
Year FE	✓		
Sample	1988–2014, excluding dissolution year		
Observations	217,984		

Notes: The table reports municipality and year fixed-effects estimates. Standard errors are clustered by municipality. The administrator-turnover coefficient is expressed in percentage points. Other coefficients are in levels. The preferred sample excludes the less stable early years 1985–1987 and excludes the dissolution year itself.

These estimates are not small. The decline in administrators is driven primarily by the decline in councillors. This is consistent with the institutional nature of dissolution, which directly interrupts elected local government. But the result is still substantively important because councillors and administrators are precisely the actors who connect local society to formal political participation. Their contraction implies a weakening of the local political infrastructure through which turnout is mobilised.

8.2 Turnover and reshuffling

The contraction in the number of administrators is accompanied by reshuffling. Administrator turnover rises by 7.26 percentage points in the preferred specification. In the baseline all-years specification, the increase is even larger, at 13.96 percentage points. Excluding the dissolution year reduces the magnitude but does not eliminate the effect. This suggests that the reshuffling is not only a mechanical feature of the dissolution year itself.

The number of distinct professions among administrators also falls. This result should be interpreted cautiously because profession coding may vary across years, but it is consistent with a narrowing or reorganisation of the local political class. The female-share and age results are less central. Female share declines modestly in the all-years specification but becomes insignificant in the preferred post-1988 sample. Mean age declines by about 1.4 years, suggesting some compositional replacement, but the core mechanism is the contraction and turnover of the local political class.

8.3 Robustness and timing

Table 13 summarises robustness checks for the main political-supply outcomes. The results are stable across samples. Excluding 1985–1987 strengthens the contraction estimates. Excluding the dissolution year reduces the magnitudes, as expected, but the main outcomes remain negative and statistically significant. Restricting to municipalities observed in at least ten snapshots leaves the baseline results essentially unchanged.

Table 13: Political-supply robustness

Sample	Administrators	Councillors	Administrator turnover	Distinct professions
All years	-5.004^{***}	-5.099^{***}	13.962^{***}	-1.538^{***}
Exclude 1985–1987	-6.509^{***}	-5.991^{***}	14.779^{***}	-1.957^{***}
Exclude dissolution year	-3.340^{***}	-3.769^{***}	6.727^{***}	-0.512^{**}
Post-1988, exclude dissolution year	-4.858^{***}	-4.662^{***}	7.264^{***}	-0.901^{***}
Municipalities with 10+ snapshots	-5.004^{***}	-5.098^{***}	13.983^{***}	-1.537^{***}

Notes: Coefficients come from municipality and year fixed-effects specifications. Administrator turnover is expressed in percentage points. Other outcomes are in levels. Standard errors are clustered by municipality. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Event-time diagnostics show very large effects in the dissolution year and the first lag, especially for administrators and councillors. This is expected because dissolution directly interrupts elected local administration. Some pre-treatment coefficients are significant, so we do not present the event study as the primary causal evidence. Its value is timing: the political-supply shock is largest around dissolution and partly attenuates over time, while some residual differences persist in later lags.

8.4 Persistence beyond the mechanical intervention year

The main threat to the political-supply interpretation is mechanical: dissolution directly removes elected authorities. The preferred specification already excludes the dissolution year, but the persistence-bin specification in Table 14 goes further by separating the immediate post-dissolution period from medium-run and longer-run effects. The table is central because it directly answers the concern that the political-supply result is merely the legal mechanics of commissioner administration. If effects were purely mechanical, the contraction should be concentrated only in the immediate period after dissolution. Instead, administrators and councillors remain significantly lower even in the 4–6 and 7+ year windows, although the magnitudes are much smaller than in years 1–3.

Table 14: Persistence of political-supply effects after the dissolution year

Post-dissolution window	Administrators	Councillors	Executive members	Administrator turnover	Distinct professions
1–3 years after dissolution	-10.491^{***}	-9.149^{***}	-2.002^{***}	19.507^{***}	-4.265^{***}
	(0.479)	(0.420)	(0.117)	(1.352)	(0.265)
4–6 years after dissolution	-1.700^{***}	-2.099^{***}	0.072	−1.298	0.709^{**}
	(0.459)	(0.389)	(0.141)	(1.803)	(0.301)
7+ years after dissolution	-3.028^{***}	-3.222^{***}	-0.239^{**}	3.829^{***}	0.307
	(0.430)	(0.396)	(0.116)	(1.266)	(0.256)
Treated municipalities, years 1–3	170	170	170	170	170
Treated municipalities, years 4–6	147	147	147	147	147
Treated municipalities, years 7+	137	137	137	137	137
Observations	217,984	217,984	217,984	215,840	217,984
Municipality FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
Sample	1988–2014, excluding dissolution year				

Notes: Municipality and year FE. Dissolution year excluded. Turnover in pp. SE clustered by municipality. $^{***}p < 0.01$, ** $^{*}p < 0.1$.

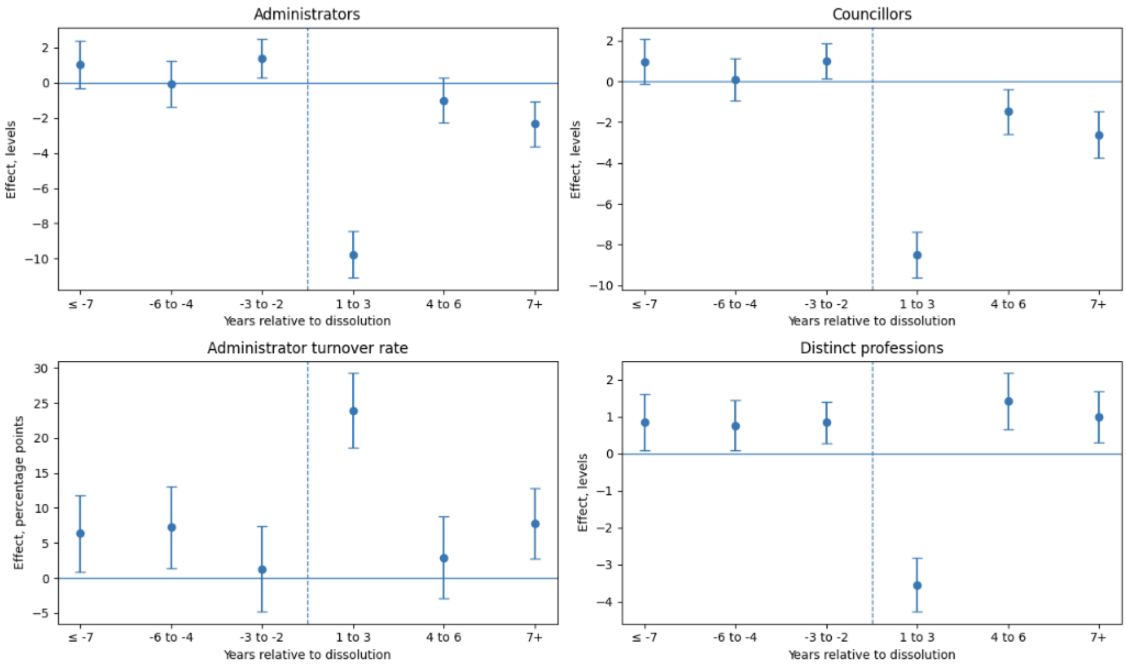

Figure 1: Political-supply event-time diagnostics

Notes: The figure reports binned event-time coefficients from municipality and year fixed-effects specifications. The omitted category is the year immediately before dissolution. The dissolution year is excluded from this diagnostic. Confidence intervals are 95 percent intervals based on standard errors clustered by municipality.

The persistence table and the event-time figure should be read together. The 1–3 year coefficients show the sharp institutional interruption expected immediately after dissolution: administrators fall by 10.5 and councillors by 9.1, while administrator turnover rises by 19.5 percentage points. The medium-run and longer-run bins are smaller, but administrators and councillors remain below their counterfactual levels in both the 4–6 and 7+ year windows. This pattern addresses the mechanical-intervention concern. The largest effect occurs immediately after dissolution, as expected, but the contraction of councillors and administrators does not disappear once the dissolution year is excluded.

The political-supply results help explain the broad turnout decline. If dissolution reduces and reshuffles the local political class, then citizens may face fewer mobilisation channels across many electoral contexts. This mechanism is not referendum-specific. It is compatible with the finding that turnout falls in ordinary elections as well as in institutional referenda.

9 Credit-Market Spillovers: SAFE–ORBIS Evidence

9.1 Short-run credit-condition tightening

The SAFE–ORBIS results show that mafia-related dissolution is associated with short-run deterioration in firms' reported credit conditions. We interpret this evidence as credit-market spillovers from the same institutional shock, not as a causal channel that mechanically explains the turnout results. Table 15 reports the main SAFE outcomes, the non-mafia placebo, and the mafia versus non-mafia horse race. The supported results are concentrated in credit-market conditions. In the baseline mafia specification, exposed province-years are 2.82 percentage points more likely to report worsened bank-loan availability, 5.03 percentage points more likely to report low confidence talking with banks, 5.18 percentage points more likely to report higher interest rates, and 2.88 percentage points more likely to report shorter loan maturity. This interpretation is consistent with evidence that local institutional quality and enforcement conditions affect credit markets (Jappelli et al., 2005).

These effects are economically meaningful and survive controls for contemporaneous firm fundamentals. After adding controls for turnover, profit, employment, and investment deterioration, the coefficient on worsened bank-loan availability rises to 3.21 percentage points, low confidence in banks rises to 5.40 percentage points, higher interest rates rise to 6.64 percentage points, and shorter maturity rises to 3.34 percentage points. This pattern suggests that the results are not simply driven by treated firms reporting worse real performance after dissolution exposure.

Table 15: SAFE–ORBIS credit conditions, non-mafia placebo, and horse race

Outcome	Mafia post 0–3	Mafia post 0–3 + Q2 controls	Non-mafia post 0–3	Horse race: mafia	Horse race: non-mafia
Discouraged borrower	-1.22	-1.43	0.54	-1.20	0.53
	(1.10)	(1.08)	(0.66)	(1.08)	(0.65)
Bank-loan availability worsened	2.82^{**}	3.21^{**}	0.51	2.84^{**}	0.54
	(1.33)	(1.43)	(1.08)	(1.30)	(1.09)
Credit-line availability worsened	2.01	2.47^{*}	0.53	2.02	0.55
	(1.25)	(1.37)	(1.38)	(1.24)	(1.39)
Not confident in banks	5.03^{**}	5.40^{**}	0.42	5.05^{**}	0.47
	(2.15)	(2.22)	(1.71)	(2.14)	(1.67)
Interest rate increased	5.18^{**}	6.64^{***}	-3.16	5.10^{**}	-3.12
	(2.21)	(2.04)	(3.01)	(2.14)	(3.06)
Loan size decreased	2.26	3.09	-0.18	2.26	-0.16
	(1.93)	(2.13)	(1.57)	(1.94)	(1.58)
Loan maturity decreased	2.88^{***}	3.34^{***}	-0.93	2.86^{***}	-0.90
	(1.11)	(1.25)	(1.08)	(1.07)	(1.07)
Collateral requirements increased	-2.20	-0.99	1.49	-2.16	1.47
	(1.72)	(1.83)	(2.00)	(1.73)	(2.00)
Bank loan rejected	-0.22	-0.06	0.53	-0.20	0.53
	(1.35)	(1.46)	(1.32)	(1.36)	(1.32)
Bank loan rejected or too costly	-0.38	-0.18	0.93	-0.35	0.92
	(1.90)	(2.02)	(1.41)	(1.95)	(1.41)

Notes: Coefficients are expressed in percentage points. Standard errors, clustered at the NUTS3 province level, are in parenthes one in the first three years after the first mafia-related municipal dissolution in the province. Non-mafia post 0–3 equals one in the first ordinary non-mafia municipal dissolution in the province. All specifications include province and year fixed effects anc Q2 controls add contemporaneous SAFE indicators for turnover, profit, employment, and investment deterioration. The horse-r mafia and non-mafia exposure in the same regression. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

9.2 Credit-vulnerability diagnostics

To evaluate whether the SAFE–ORBIS effects are driven by weak-firm composition, we split firms by predicted credit vulnerability and also estimate risk-gradient specifications. The results support the intensive-margin interpretation. Worsened bank-loan availability increases by 3.2 percentage points for firms at median predicted availability vulnerability, and the interaction with predicted vulnerability is not statistically significant. This suggests that the bank-loan availability effect is not confined to firms that were already highly vulnerable. By contrast, credit-line availability, interest-rate increases, and loan-size reductions are more concentrated among high-vulnerability firms. In the median-split specification, high price-and-term vulnerability firms are 15.5 percentage points more likely to report higher interest rates, while low-vulnerability firms do not report a comparable increase. Hard-rationing outcomes remain weak: discouraged borrowing and rejection do not display a robust positive pattern. Overall, the credit-vulnerability diagnostics indicate that mafia-related dissolution generates a broad deterioration in perceived bank-loan availability, combined with amplified screening of already vulnerable firms on price and credit-line margins.

9.3 Non-mafia placebo and mafia versus non-mafia horse race

The non-mafia placebo is central for interpretation. Ordinary municipal dissolutions are also disruptive administrative events, but they do not publicly certify mafia infiltration. If the SAFE effects were driven by generic local administrative instability, the non-mafia placebo should generate similar credit tightening. It does not. Across all ten outcomes in Table 15, the non-mafia post 0–3 coefficient is small and statistically insignificant.

The horse-race specification confirms this pattern. When mafia-related and ordinary non-mafia dissolutions are included together, the mafia coefficient remains almost unchanged for the key credit outcomes. Bank-loan availability worsens by 2.84 percentage points, low confidence in banks rises by 5.05 percentage points, interest-rate increases rise by 5.10 percentage points, and maturity deterioration rises by 2.86 percentage points. The corresponding non-mafia coefficients remain close to zero and insignificant. With Q2 controls, the mafia effects become 3.23, 5.41, 6.57, and 3.33 percentage points respectively, while the non-mafia coefficients remain null.

This evidence strengthens the institutional-disruption interpretation. The SAFE results are not a general response to council dissolution or administrative instability. They are specific to the kind of dissolution that reveals mafia-related institutional contamination. This supports the view that banks and firms react not only to legal intervention, but to the information that local institutions and networks may have been penetrated by organised crime.

9.4 Null margins, timing, and interpretation

The SAFE results also narrow what the paper can claim. Discouraged borrowing does not increase. Bank-loan rejection does not increase. Bank loans rejected or judged too costly do not increase. Collateral requirements do not rise, and loan-size reductions are positive but not statistically significant. The supported mechanism is therefore not a broad exclusion story in which firms leave the credit market or banks reject applications more frequently. It is an intensive-margin tightening story: firms report worse availability, higher interest rates, shorter maturities, and lower confidence in banks.

The null effects on discouraged borrowing and rejection are power-limited rather than decisive evidence of no response. The baseline rates of discouraged borrowing and bank-loan rejection are low, around four to five percent, and the treated support in the short-run province-year window is limited. The point estimates are small and statistically insignificant, so the data do not support a discouraged-borrower or rejection mechanism. At the same time, these outcomes should be interpreted as low-power nulls rather than as strong evidence that extensive-margin credit rationing is absent.

Timing diagnostics indicate that the credit effects are strongest in the first years after exposure and then become less precise. This is consistent with a short-run disruption rather than a permanent credit scar. The SAFE evidence has limits. Placebo leads are not perfectly clean, and province-level exposure may capture broader local dynamics. Because firm geography is observed at NUTS3 level and SAFE is selective, these results should be interpreted as credit-market spillover evidence rather than as proof of a separate municipality-level causal channel. The key contribution of SAFE is to show that the institutional shock is accompanied by measurable disruption in local credit relationships, and that this disruption is not reproduced by ordinary non-mafia dissolutions.

10 Discussion

10.1 One shock, two infrastructures

The results point to one institutional shock affecting two infrastructures. The first is democratic infrastructure. Mafia-related dissolution contracts and reshuffles the local political class. Administrators and councillors decline, and turnover increases. These changes plausibly weaken local mobilisation capacity and help explain the broad turnout decline across elections.

The second is economic infrastructure. SAFE–ORBIS evidence shows that firms in exposed provinces report worse bank-loan availability, lower confidence in banks, higher interest rates, and shorter maturities. Ordinary non-mafia dissolutions do not generate the same effects. This means that the economic disruption is not simply a response to generic administrative instability. It is specifically associated with the revelation of mafia-related institutional contamination.

10.2 Why credit-market spillovers matter

SAFE evidence should not be used to claim that credit tightening causes turnout decline. The timing is too short-run, and the geography is too coarse. Its value is different. It shows that the same institutional shock that weakens local democratic participation is also associated with deterioration in local credit relationships. In this sense, SAFE provides spillover evidence: mafia-related dissolution is visible beyond the ballot box and beyond municipal offices.

This distinction is important. The paper does not require credit tightening to explain electoral demobilisation. Instead, the political-supply results provide the main mechanism for the participation scar, while the SAFE evidence shows that dissolution also unsettles local economic relationships. The non-mafia placebo makes this interpretation more specific. Ordinary non-mafia dissolutions do not generate the same credit tightening, suggesting that banks and firms respond to the revelation of mafia-related institutional contamination rather than to generic administrative instability.

10.3 Policy implications

The findings do not imply that mafia-related dissolution is bad policy. Removing compromised administrations may be necessary to restore legality. The results instead highlight a democratic trade-off. Legal cleansing is not the same as democratic reconstruction. If dissolution removes or reshuffles the local political class and weakens mobilisation, the state may need to invest more actively in rebuilding local democratic capacity after the extraordinary commission period.

The policy implication is therefore straightforward: anti-mafia intervention should not end with administrative replacement. It should include measures that rebuild local participation, support credible political supply, and restore confidence among citizens and firms. Cleaning institutions is necessary, but it is not sufficient if the intervention leaves a durable participation scar.

11 Conclusion

This paper studies the consequences of mafia-related municipal dissolution in Italy. The evidence shows that dissolution leaves a political scar. Referendum turnout falls after dissolution, and broader election-family comparisons show that the participation decline is not limited to judicial or institutional referenda. The scar is broad.

The paper also shows that this broad demobilisation is accompanied by a disruption of local political supply. Annual administrator data reveal fewer administrators, fewer councillors, and higher turnover after dissolution. This mechanism is central because political participation depends not only on individual attitudes but also on the local actors who mobilise citizens and maintain democratic continuity.

Finally, SAFE–ORBIS evidence shows that dissolution is associated with short-run deterioration in firm-bank credit conditions. Firms in exposed provinces report worse bank-loan availability, lower confidence in banks, higher interest rates, and shorter maturities. Ordinary non-mafia municipal dissolutions do not reproduce these effects, and the mafia coefficients remain stable in horse-race specifications. This economic evidence is not a standalone municipality-level causal design, but it supports the broader interpretation of dissolution as a local institutional shock with credit-market spillovers.

The main conclusion is that mafia-related dissolution does more than remove compromised administrations. It disrupts local political and economic infrastructure. The intervention may be legally necessary, but it can leave a democratic scar unless accompanied by efforts to rebuild political supply, civic participation, and institutional confidence.

A Additional Tables and Diagnostics

A.1 Pre-treatment levels and event-study diagnostics

Future-treated municipalities have lower pre-treatment turnout levels than never-treated municipalities. These level differences do not by themselves invalidate the design, because the identifying assumption concerns trends rather than equality of baseline levels. The original event-study leads are not jointly significant, but the stacked and political-supply event-time diagnostics are treated cautiously because some pre-treatment differences remain.

Table 16: Pre-treatment level differences and event-study lead diagnostics

Check	Estimate	p-value	Interpretation
Pre-treatment turnout level difference, unweighted	-5.383	0.0001	Future-treated municipalities ha
Pre-treatment turnout level difference, electorate-weighted	-3.858	0.0003	Future-treated municipalities ha
Joint test of original event-study leads	$\chi^2(3) = 2.421$	0.490	No rejection of zero pre-trea

Notes: Level differences compare future-treated and never-treated municipalities before treatment. The event-study lead tes municipality and referendum-wave fixed-effects specification. The table separates baseline level differences from identifying pre-

A.2 Holm-adjusted inference for 2026 composition outcomes

Table 17 reports Holm-adjusted inference for the three primary electorate-based 2026 composition outcomes. The adjustment is applied within each weighting scheme across Yes votes as a share of the electorate, No votes as a share of the electorate, and valid votes as a share of the electorate. The decline in No votes and the decline in valid votes remain significant under all weighting schemes, while the Yes-vote outcome does not survive the correction.

Table 17: Holm-adjusted inference for 2026 electorate-based composition outcomes

Weighting	Outcome	Coefficient	Raw p-value	Holm p-value
Unweighted	Yes votes / electorate	1.094	0.083	0.083
Unweighted	No votes / electorate	-2.067	< 0.001	< 0.001
Unweighted	Valid votes / electorate	-0.972	0.005	0.014
Weighted by valid votes	Yes votes / electorate	0.105	0.745	0.745
Weighted by valid votes	No votes / electorate	-1.062	0.019	0.037
Weighted by valid votes	Valid votes / electorate	-0.957	0.002	0.006
Weighted by electorate	Yes votes / electorate	0.119	0.704	0.704
Weighted by electorate	No votes / electorate	-1.110	0.013	0.025
Weighted by electorate	Valid votes / electorate	-0.991	0.002	0.005

Notes: Holm adjustment is applied across the three primary electorate-based outcomes within each weighting scheme. Coefficients are expressed in percentage points. The result supports the status-quo demobilisation interpretation because the No-vote and valid-vote outcomes survive adjustment, while the Yes-vote outcome does not.

A.3 Modern staggered-adoption diagnostics

Because treatment timing is staggered across municipalities, we report a Callaway–Sant'Anna-style group-time ATT estimator as a robustness check for the main referendum turnout result. This estimator compares newly treated municipalities with never-treated and not-yet-treated municipalities and avoids using already-treated units as controls. The resulting estimates are smaller and less precise than the TWFE estimates. The unweighted 0–3 wave ATT is -0.68 percentage points, while the electorate-weighted 0–3 wave ATT is -2.00 percentage points. The latter is close to the province-by-wave fixed-effects estimate of -1.81 percentage points, but it is not statistically significant. These results support a cautious interpretation: the sign of the participation scar is robust in the weighted modern-DiD diagnostic, but the exact magnitude should not be over-interpreted.

A.4 Electoral-family robustness and interpretation

The electoral-family extension is diagnostic. It helps reject a narrow referendum-only interpretation, but it is not the strongest causal design because election families are observed in different years. Table 18 summarises the interpretation of the relevant estimates.

Table 18: How the electoral-family extension should be interpreted

Result	Status	Interpretation
Non-referendum turnout effect of about -11.4 pp	Diagnostic	Shows that the participation scar is not confined to referenda, but the magnitude may reflect election timing, salience, and sample composition.
2020 constitutional referendum effect of about -9.3 pp	Diagnostic and mechanism-relevant	Supports institutional demobilisation in a constitutional setting, but should not be pooled mechanically with ordinary elections.
2022 judicial referenda effect of about -4.7 pp	Weaker diagnostic	Negative but less precise. Does not support the claim that judicial referenda are uniquely or most strongly affected.
2026 No/electorate decline	Main composition evidence	Supports status-quo demobilisation in the judicial-governance referendum, especially when combined with matched Southern DiD.

Notes: The table clarifies the inferential status of the electoral-family extension. It should be used to broaden the interpretation from referendum-specific demobilisation to a broader participation scar, not as the primary causal design.

Table 19: Electoral-family robustness across comparison samples

Sample and term	Coefficient	Standard error	p-value
Baseline, non-referendum	-11.426^{***}	(2.566)	< 0.001
Baseline, constitutional referendum	-9.315^{***}	(2.823)	0.001
Baseline, judicial referenda	-4.729^{*}	(2.798)	0.091
Exclude 2020 regional elections, non-referendum	-11.460^{***}	(2.909)	< 0.001
Exclude 2020 regional elections, constitutional	-7.788^{**}	(3.175)	0.014
Exclude 2020 regional elections, judicial	-3.262	(3.128)	0.297
Exclude 2024 European election, non-referendum	-9.530^{**}	(4.341)	0.028
Exclude 2024 European election, constitutional	-8.461^{*}	(4.498)	0.060
Exclude 2024 European election, judicial	-3.785	(4.354)	0.385
Exclude 2022 political election, non-referendum	-12.477^{***}	(2.915)	< 0.001
Exclude 2022 political election, constitutional	-11.876^{***}	(3.119)	< 0.001
Exclude 2022 political election, judicial	-7.156^{**}	(3.054)	0.019
Referenda plus 2022 political only, non-referendum	-8.905^{*}	(4.714)	0.059
Referenda plus 2022 political only, constitutional	-4.774	(4.891)	0.329
Referenda plus 2022 political only, judicial	-0.183	(4.731)	0.969
Referenda plus 2024 European only, non-referendum	-13.106^{***}	(3.278)	< 0.001
Referenda plus 2024 European only, constitutional	-10.059^{***}	(3.518)	0.004
Referenda plus 2024 European only, judicial	-5.417	(3.426)	0.114

Notes: The dependent variable is turnout. Coefficients are in percentage points. Each row comes from a municipality and election fixed-effects specification with standard errors clustered by municipality. The table is used as a diagnostic because election families are observed in different years and differ in salience and campaign context.

A.5 Non-mafia dissolution placebo

The electoral analysis compares mafia-related dissolution with a non-mafia municipal-dissolution placebo. In the strict electoral panel specification, mafia-related dissolution reduces turnout by 1.81 percentage points, while the non-mafia placebo effect is small and insignificant. The difference between the two effects is statistically significant. This supports the interpretation that the electoral scar is connected to mafia-related institutional contamination rather than generic municipal instability.

For the SAFE–ORBIS analysis, we construct a separate official non-mafia placebo from INT00063 municipal dissolution files covering 2016–2024. This placebo contains 1,118 ordinary non-mafia municipal dissolutions and matches 98.2 percent of rows to SAFE province names. In the SAFE horse race, ordinary non-mafia dissolutions do not reproduce the credit-tightening effects of mafia-related dissolutions. This makes the SAFE evidence more specific: the observed credit disruption is associated with mafia-related institutional contamination, not with ordinary council dissolution per se.

A.6 Treatment intensity and regional heterogeneity

Repeated mafia-related dissolution produces larger declines in status-quo votes than one-time dissolution. Table 20 shows that repeated dissolutions are associated with a much larger fall in No votes as a share of the electorate than one-time dissolutions. This supports a dose-response interpretation in which deeper or repeated institutional disruption leaves stronger political scars.

Table 20: Treatment intensity and status-quo demobilisation

	No votes / electorate
One mafia-related dissolution	-1.424
Repeated mafia-related dissolutions	-4.010^{***}
Difference: repeated − one-time	-2.587^{***}
p-value of difference	0.0005

Notes: The outcome is No votes as a share of the electorate in the 2026 judicial-governance referendum. The estimates distinguish municipalities with one pre-referendum mafia-related dissolution from municipalities with repeated pre-referendum dissolutions. The difference test evaluates whether the repeated-dissolution coefficient is more negative than the one-time-dissolution coefficient. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

The composition effect is also stronger in the Calabria/Sicily group than in the Campania/Puglia group, especially for No votes as a share of the electorate. Table 21 reports the formal heterogeneity tests. These tests are diagnostic rather than a separate causal design, but they support the interpretation that the political scar is larger where mafia-related institutional embeddedness is plausibly deeper.

Table 21: Formal regional heterogeneity tests by mafia typology

Contrast and outcome	Difference	p-value	FDR-adjusted p-value
Calabria − Campania, No votes / electorate	-2.23	0.006	0.011
Calabria − Puglia, No votes / electorate	-2.42	0.002	0.010
Calabria/Sicily average − Campania/Puglia average, No votes / electorate	-1.54	0.004	0.011
Calabria/Sicily average − Campania/Puglia average, Yes share / valid votes	2.45	0.005	0.013
Calabria/Sicily average − Campania/Puglia average, turnout	-1.01	0.022	0.096

Notes: The table reports formal difference tests from the South-only 2026 typology specification. Calabria and Sicily are grouped as regions associated with deeper institutional embeddedness of mafia organisations, while Campania and Puglia are grouped as regions associated with more fragmented or less territorially embedded mafia structures. The tests are heterogeneity diagnostics, not a separate causal design. Coefficients are expressed in percentage points.

A.7 Political-supply coverage and robustness

The political-supply panel is built from annual snapshots of local administrators. The corrected parser reads the semicolon-separated files and produces plausible municipality-year coverage. Coverage is less stable in 1985–1987, which motivates the preferred post-1988 sample. Table 22 reports the core coverage diagnostics.

Table 22: Political-supply data coverage

Coverage statistic	Value
Years covered	1985–2014
Preferred years	1988–2014
Total municipality-year observations	237,897
Matched treated municipalities	271
Treatment municipalities in treatment data	274
Preferred sample observations	217,984
Main reason for excluding 1985–1987	Less stable early coverage
Main reason for excluding dissolution year	Avoid mechanical commissioner-year effects

Notes: The preferred sample used in the main political-supply table excludes 1985–1987 and the dissolution year. This is the main submission specification because it addresses both coverage instability and mechanical intervention-year effects.

Table 23: Political-supply robustness summary

Sample	Administrators	Councillors	Administrator turnover	Distinct professions
All years	-5.004^{***}	-5.099^{***}	13.962^{***}	-1.538^{***}
Exclude 1985–1987	-6.509^{***}	-5.991^{***}	14.779^{***}	-1.957^{***}
Exclude dissolution year	-3.340^{***}	-3.769^{***}	6.727^{***}	-0.512^{**}
Post-1988, exclude dissolution year	-4.858^{***}	-4.662^{***}	7.264^{***}	-0.901^{***}
Municipalities with 10+ snapshots	-5.004^{***}	-5.098^{***}	13.983^{***}	-1.537^{***}

Notes: Coefficients come from municipality and year fixed-effects specifications. Administrator turnover is expressed in percentage points. Other outcomes are in levels. Standard errors are clustered by municipality. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

A.8 SAFE–ORBIS diagnostics

The SAFE–ORBIS main text reports the selected credit-condition, confidence, price-term, and placebo horse-race outcomes. This appendix documents the broader diagnostics needed for a submission version: variable construction, attribution diagnostics, real-performance outcomes, timing, attrition, firm fixed effects, and Q32 reason variables.

Table 24: Construction of SAFE dependent variables

Constructed outcome	SAFE variable	Coding rule
Bank-loan need increased	`q5_a_rec`	1 if response indicates increased need
Worsened bank-loan availability	`q9_a_rec`	1 if response indicates deteriorated availability
Available loan size decreased	`q10_c_rec`	1 if response indicates decreased available size
Loan maturity decreased	`q10_d_rec`	1 if response indicates decreased maturity
Expected worsening in bank-loan availability	`q23_b_rec`	1 if response indicates expected deterioration
Interest rate increased	`q10_a_rec`	1 if response indicates increased interest rate
Not confident talking with banks	`q19_a`	1 if response is No
Discouraged bank-loan borrower	`q7a_a_rec`	1 if firm did not apply because of possible rejection

Notes: Non-applicable, refusal, and don't-know responses are treated as missing. For q19, code 1 means Yes and code 2 means No. The dependent variable for bank confidence is coded as one when the firm answers No to being confident talking about financing with banks. For q32, code 8 is treated as a valid "no need" category rather than as missing.

Attribution diagnostics. The original roadmap considered SAFE attribution variables as a potential way to distinguish institutional credit-market concerns from firm-specific deterioration. This test is not supportive. Internal attribution items move more clearly than external attribution items, and the external-versus-internal contrast is not statistically significant. We therefore do not use the attribution battery as a central mechanism test. Instead, the main defence against a pure firm-fundamentals interpretation comes from the Q2-control robustness, where the credit-condition results survive controls for contemporaneous turnover, profit, employment, and investment deterioration.

Real-performance outcomes. The SAFE real-performance outcomes do not support the hypothesis that treated firms experience a contemporaneous deterioration in reported turnover or profits. If anything, some estimates move in the opposite direction. We therefore avoid interpreting the credit-condition results as mechanically driven by weaker reported firm performance. One possible explanation is survey selection among surviving firms. Another possibility is that dissolution weakens captured local networks and may benefit some legitimate firms. The data do not allow me to distinguish these interpretations directly, so the real-performance results are treated as a limitation and not as a separate mechanism.

Table 25: Timing of selected SAFE credit outcomes

Outcome and event-time bin	Coefficient	Standard error	p-value
Worsened bank-loan availability, 0 to 1 years	0.084	0.031	0.007
Worsened bank-loan availability, 2 to 3 years	0.013	0.031	0.691
Worsened bank-loan availability, 4 to 5 years	0.021	0.043	0.620
Worsened bank-loan availability, 6+ years	0.020	0.036	0.589
Available loan size decreased, 0 to 1 years	0.095	0.038	0.013
Available loan size decreased, 2 to 3 years	0.029	0.036	0.414
Available loan size decreased, 4 to 5 years	0.042	0.061	0.492
Available loan size decreased, 6+ years	0.030	0.049	0.542
Loan maturity decreased, 0 to 1 years	-0.014	0.039	0.720
Loan maturity decreased, 2 to 3 years	0.018	0.032	0.561
Loan maturity decreased, 4 to 5 years	-0.016	0.034	0.644
Loan maturity decreased, 6+ years	-0.028	0.034	0.409

Notes: The omitted category is the year immediately before exposure. Regressions include province and year fixed effects and lagged ORBIS controls. Standard errors are clustered at NUTS3 level. The timing estimates are diagnostic because broader placebo-lead specifications are not perfectly clean.

Table 26: SAFE panel attrition

Sample and treatment window	Coefficient	Standard error	p-value
All Italy, 0 to 3 years	0.032	0.025	0.198
All Italy, 0 to 5 years	0.019	0.031	0.538
All Italy, 6+ years	-0.027	0.023	0.238
South, 0 to 3 years	-0.066	0.090	0.467
South, 0 to 5 years	0.081	0.060	0.176
South, 6+ years	-0.064	0.086	0.461

Notes: The dependent variable equals one if the firm is observed again in SAFE in the following year. The all-Italy regressions include province and year fixed effects. The South-only regressions include province and region-year fixed effects. All specifications include lagged ORBIS controls and cluster standard errors at NUTS3 level.

Table 27: SAFE firm fixed-effects robustness

Outcome	Coefficient	Standard error	p-value
Bank-loan need increased	0.040	0.022	0.073
Worsened bank-loan availability	0.047	0.019	0.012
Available loan size decreased	0.067	0.026	0.009
Loan maturity decreased	0.028	0.012	0.021
Expected worsening in bank-loan availability	0.040	0.018	0.025

Notes: The sample is restricted to firms observed in at least two SAFE years. Regressions include firm and year fixed effects, lagged ORBIS controls, SAFE weights, and standard errors clustered at NUTS3 level.

Table 28: Corrected Q32 SAFE reason diagnostics

Outcome	Coefficient	p-value	Interpretation
Bank loans not relevant because of supply/friction reasons	-0.015	0.618	No evidence of a shift toward supply or friction reasons
Bank loans not relevant because of no need	-0.005	0.814	No meaningful change in “no need” responses

Notes: These outcomes use corrected Q32 coding in which code 8 is preserved as the valid “no need” category. Supply/friction reasons include responses related to collateral, interest rates, lack of availability, or paperwork. The estimates come from the province-year roadmap specification with province and year fixed effects, lagged ORBIS controls, SAFE weights, and standard errors clustered at NUTS3 level.

Table 29: SAFE credit outcomes by predicted credit vulnerability

Credit-vulnerability dimension	Outcome	Low vulnerability	High vulnerability	Low – High	p-value diff.	Obs. / Clusters
		2.75***	3.71	-0.96		
Availability vulnerability	Bank-loan availability worsened	(0.98)	(2.40)	(2.65)	0.717	15,387 / 104
		0.90	5.93***	-5.03**		
Availability vulnerability	Credit-line availability worsened	(1.17)	(2.24)	(2.26)	0.026	12,320 / 104
		1.41	8.42***	-7.01		
Bank-confidence vulnerability	Not confident talking with banks	(2.42)	(3.09)	(4.71)	0.137	10,848 / 105
		-1.38	15.54***	-16.92***		
Price/term vulnerability	Interest rate increased	(4.79)	(3.49)	(6.26)	0.007	7,815 / 104
		0.25	5.09**	-4.84**		
Price/term vulnerability	Available loan size decreased	(0.92)	(2.57)	(2.27)	0.033	7,821 / 104
		4.18***	2.07	2.11		
Price/term vulnerability	Loan maturity decreased	(0.97)	(1.80)	(1.29)	0.102	7,737 / 104
		-3.07*	2.77	-5.84**		
Price/term vulnerability	Collateral requirements increased	(1.74)	(2.75)	(2.88)	0.043	7,551 / 104
		-1.59**	-0.52	-1.07		
Hard-rationing vulnerability	Discouraged borrower	(0.76)	(1.92)	(2.35)	0.649	15,901 / 104
		1.87***	-0.22	2.09		
Hard-rationing vulnerability	Bank-loan rejected	(0.50)	(1.30)	(1.31)	0.110	5,137 / 104
		1.46	-0.50	1.97		
Hard-rationing vulnerability	Rejected or too costly	(0.97)	(2.08)	(1.80)	0.275	5,137 / 104

Notes: Coefficients are expressed in percentage points. Low and high vulnerability are defined relative to the median predicted probability of the relevant credit-vulnerability dimension. Predicted probabilities are estimated using never-exposed and pre-exposure firm-wave observations and lagged ORBIS fundamentals. The four vulnerability dimensions are availability vulnerability, price/term vulnerability, hard-rationing vulnerability, and bank-confidence vulnerability. All regressions include province fixed effects, year fixed effects, SAFE weights, and lagged ORBIS controls. Standard errors, clustered at the NUTS3 province level, are reported in parentheses. The column “Low – High” tests whether the post-exposure effect differs between low- and high-vulnerability firms. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

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